The Role of Consolidation and Relearning in Unconscious Priming From Newly-Acquired Category Exemplars

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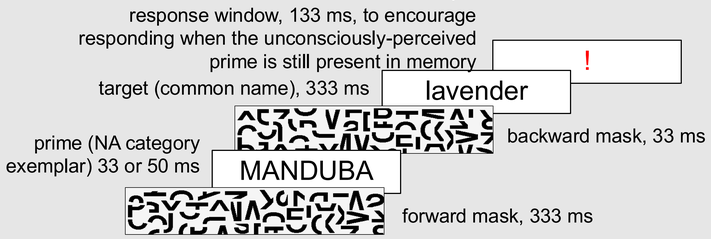
Introduction

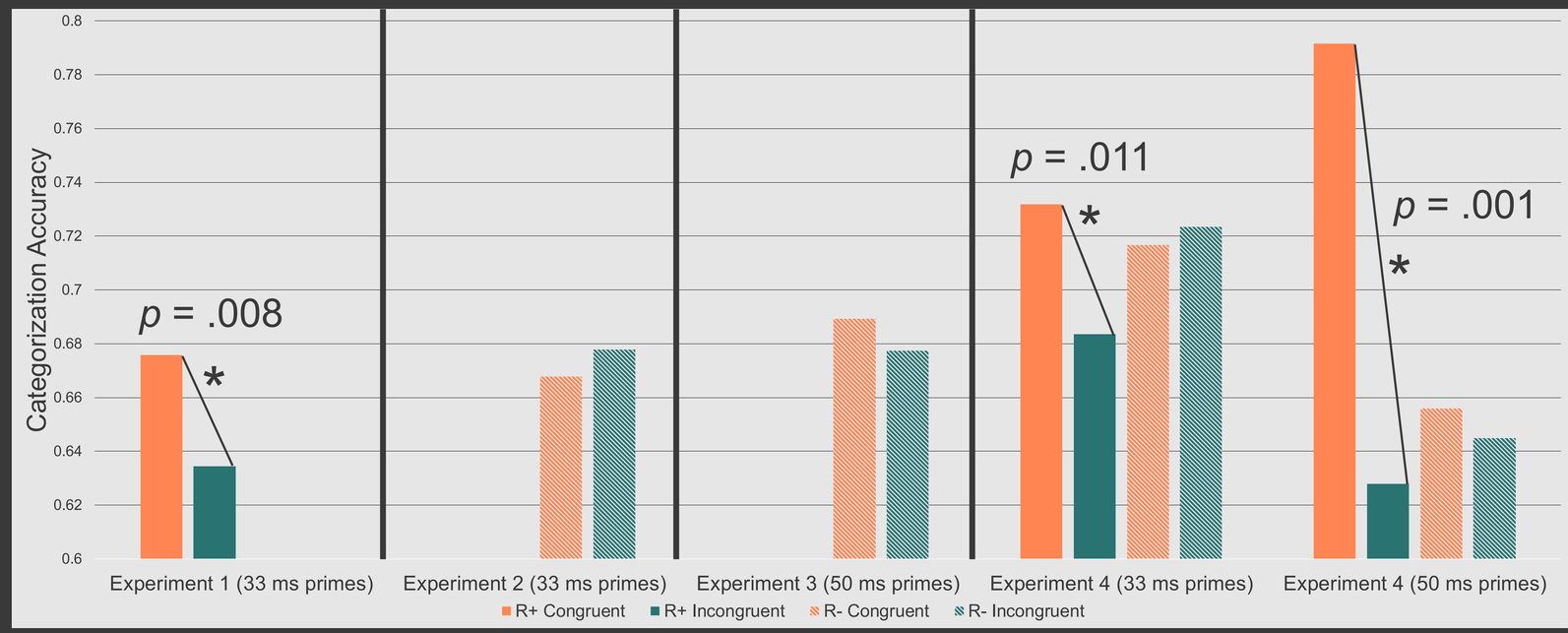
- Unconscious semantic priming is a phenomenon in which a briefly flashed, consciously imperceptible word influences the categorization of a different target word that participants are instructed to attend to.
- Unconscious semantic priming research suggests that, in order to be successful, primes must be well known (overlearned) words and must be "relearned," or practiced consciously before their use as unconscious primes (Damian, 2001; Greenwald & Abrams, 2000).
- Some research reports that after extensive learning, novel words successfully acted as semantic primes, though whether their processing was unconscious or not is uncertain (Dagenbach, Horst, & Carr, 1990; Tamminen & Gaskell, 2013).
- Our studies focused on the recruitment of newly acquired words under display conditions typically associated with unconscious processing.
- Over the course of four experiments, the acquisition of rare and presumed-to-be unfamiliar exemplars of two common categories, fish and flower, was examined. We manipulated (a) the duration between practice and priming task sessions, (b) whether or not participants received a brief second round of practice (relearning) immediately before priming, and (c) prime duration. Our objective was to evaluate the process by which newly-acquired vocabulary is recruited into unconscious processing and to examine the role of relearning.

Method

- All experiments began with a session of extensive learning (30-45 minutes) in which participants were introduced to the novel exemplars first by pairing the words with pictures of the corresponding fish or flower, then practiced categorizing the words alone.
- After a time-varied interval, participants performed a priming task with the newly-acquired (NA) exemplars as the masked primes, and common fish and flower names as the visible targets.
- A relearning session with the newly-acquired exemplars just before the priming task was given to all participants in Experiment 1, none in Experiments 2 and 3, and to half in Experiment 4. Priming tasks were followed by a test of prime visibility in Experiments 2-4, in which participants attempted to categorize the briefly-flashed, visually masked primes.

Trial sequence in priming and visibility tasks
(Same sequence in both tasks; in priming, instruction was to categorize the target; in visibility, instruction was to categorize the masked word)





Results

Experiment 1 (*N* = 19; relearning [R+] after a 24-hour interval, 33 ms primes)

- NA exemplars yielded significant priming (p = .008), operationalized as a significant difference in accuracy between congruent and incongruent trials in which the NA exemplars were primes.

Experient 2 (*N* = 42; no relearning [R-] after a 0-hour interval, 33 ms primes)

- NA exemplars did not yield significant priming, p = .270.

Experiment 3 (*N* = 42; no relearning [R-] after a 4-hour interval, 50 ms primes)

- NA exemplars did not yield significant priming, p = .244.

Experiment 4 (*N* = 47; relearning [R+] or no relearning [R-] after an 18-hour interval, 33ms or 50 ms primes between-Ss)

- NA exemplars yielded significant priming for both R+ groups, 33 ms (n = 12; p = .011) and 50 ms (n = 11, p = .001).
- No significant priming from the NA exemplars in either R- group, 33 ms (n = 11; p = .753) nor 50 ms (n = 13; p = .528).

General

- Priming was significant when common exemplars were used as primes in all four experiments, ps < 05.
- Prime visibility did not differ from chance (0.5) for both the common and NA exemplars in Experiments 2-4 (not tested in Experiment 1), ps > .05.

Discussion

- These experiments demonstrate that novel category exemplars are rapidly recruited into systems that handle unconscious processing of visual language.
- Similarly to overlearned words, these NA category exemplars function as unconscious primes only after a session of relearning.
- Prime duration influences the size of the priming effect, with longer prime durations yielding more priming.
- The novel category exemplars' ability to function as unconscious primes emerges within 18 hours and lasts for at least 24 hours.
- Successful priming after 18 hours suggests that rapid recruitment may depend upon a prolonged opportunity to consolidate the newly learned verbal material, such as the intensive consolidation that occurs during overnight sleep (e.g. Saletin & Walker, 2012). This opens up the intriguing possibility that that overnight sleep is necessary for the recruitment of novel words into unconscious processing systems.

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